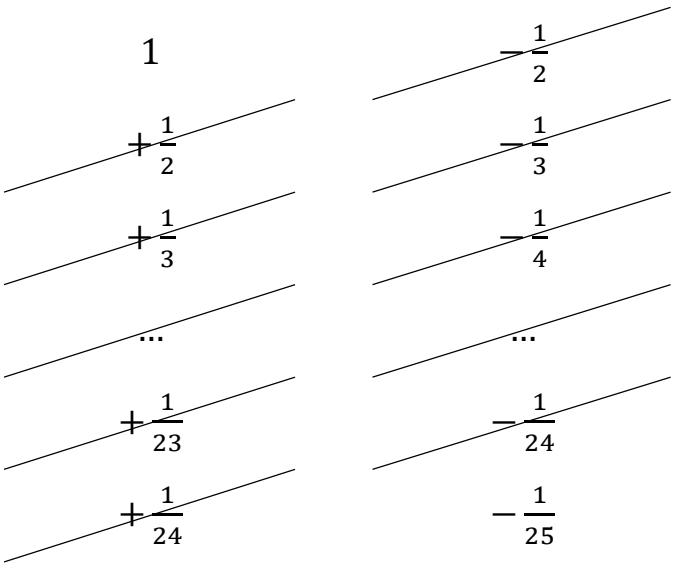
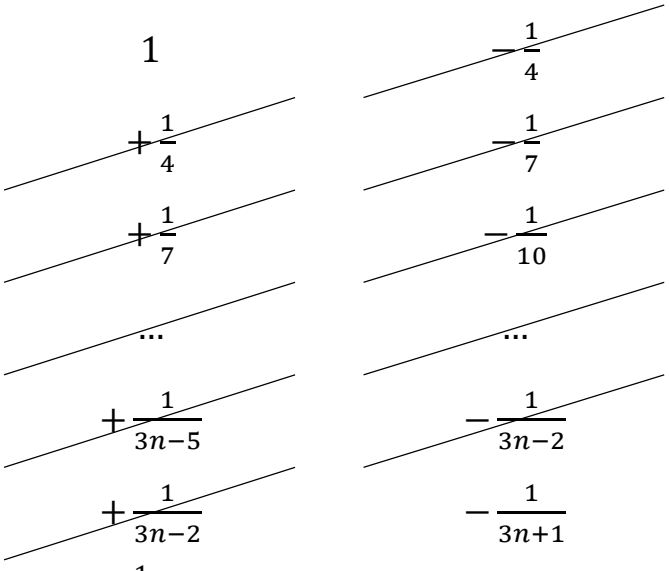
**1) a)**




$= 1 - \frac{1}{25} = \frac{24}{25}$

**1) b)**




$= 1 - \frac{1}{3n+1}$

**1) c)**

$$\begin{array}{ll} \frac{1}{6} & -\frac{1}{10} \\ +\frac{1}{8} & -\frac{1}{12} \\ +\frac{1}{10} & -\frac{1}{14} \\ +\frac{1}{12} & -\frac{1}{16} \\ \ldots & \ldots \\ +\frac{1}{2n} & -\frac{1}{2(n+2)} \\ +\frac{1}{2(n+1)} & -\frac{1}{2(n+3)} \\ +\frac{1}{2(n+2)} & -\frac{1}{2(n+4)} \end{array}$$

$$= \frac{1}{6} + \frac{1}{8} - \frac{1}{2(n+3)} - \frac{1}{2(n+4)} = \frac{7}{24} - \frac{1}{2(n+3)} - \frac{1}{2(n+4)}$$

**1) d)**

$$\begin{array}{ll} 2 & -1 \\ +3 & -2 \\ +4 & -3 \\ \ldots & \ldots \\ +(n-1) & -(n-2) \\ +n & -(n-1) \\ +(n+1) & -n \end{array}$$

$$= -1 + (n+1) = n$$

**1) e)**

$1 \times 2$ $\quad -1 \times 0$

$+2 \times 3$ $\quad -2 \times 1$

$+3 \times 4$ $\quad -3 \times 2$

$+4 \times 5$ $\quad -4 \times 3$

... $\quad$ ...

$+(n-1)n$ $\quad -(n-1)(n-2)$

$+n(n+1)$ $\quad -n(n-1)$

$= -0 + n(n+1) = n(n+1)$

**2) a)**

$$\frac{1}{n+1} - \frac{1}{n+2} = \frac{n+2}{(n+1)(n+2)} - \frac{n+1}{(n+1)(n+2)} = \frac{n+2-n-1}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}$$

**2) b)**

$$1 + \frac{1}{n+1} = \frac{n+1}{n+1} + \frac{1}{n+1} = \frac{n+2}{n+1}$$

**2) c)**

$$\frac{1}{2} - \frac{1}{n+2} = \frac{n+2}{2(n+2)} - \frac{2}{2(n+2)} = \frac{n+2-2}{2(n+2)} = \frac{n}{2(n+2)}$$

**2) d)**

$$-\frac{3}{2} + \frac{1}{n+2} + \frac{1}{n+1} = -\frac{3(n+1)(n+2)}{2(n+1)(n+2)} + \frac{2(n+1)}{2(n+1)(n+2)} + \frac{2(n+2)}{2(n+1)(n+2)}$$

$$= -\frac{3n^2+9n+2}{2(n+1)(n+2)} + \frac{2n+2}{2(n+1)(n+2)} + \frac{2n+4}{2(n+1)(n+2)}$$

$$= \frac{-3n^2-9n-2+2n+2+2n+4}{2(n+1)(n+2)}$$

$$= \frac{-3n^2-5n}{2(n+1)(n+2)}$$

$$= \frac{-n(3n+5)}{2(n+1)(n+2)}$$

**2) e)**

$$\frac{1}{12}+\frac{1}{2(n+3)}-\frac{1}{2(n+2)}=\frac{(n+2)(n+3)}{12(n+2)(n+3)}+\frac{6(n+2)}{12(n+2)(n+3)}-\frac{6(n+3)}{12(n+2)(n+3)}$$

$$=\frac{n^2+5n+6}{12(n+2)(n+3)}+\frac{6n+12}{12(n+2)(n+3)}-\frac{6n+18}{12(n+2)(n+3)}$$

$$=\frac{n^2+5n+6+6n+12-6n-18}{12(n+2)(n+3)}$$

$$=\frac{n^2+5n}{12(n+2)(n+3)}$$

**3)**

$$\sum_{r=1}^{50}\left(\frac{1}{r+2}-\frac{1}{r}\right)=$$

$$\begin{array}{rr}
\frac{1}{3} & -\frac{1}{1} \\
+\frac{1}{4} & -\frac{1}{2} \\
+\frac{1}{5} & -\frac{1}{3} \\
+\frac{1}{6} & -\frac{1}{4} \\
\ldots & \ldots \\
+\frac{1}{50} & -\frac{1}{48} \\
+\frac{1}{51} & -\frac{1}{49} \\
+\frac{1}{52} & -\frac{1}{50}
\end{array}$$

$$=-1-\frac{1}{2}+\frac{1}{51}+\frac{1}{52}=-\frac{3875}{2652}$$

**4)**

$$\sum_{r=8}^{100}\left(\frac{1}{r+1}-\frac{1}{r}\right)=$$

$$\frac{1}{9}+\frac{1}{10}+\frac{1}{11}+\cdots+\frac{1}{99}+\frac{1}{100}+\frac{1}{101}$$

$$-\frac{1}{8}-\frac{1}{9}-\frac{1}{10}-\cdots-\frac{1}{98}-\frac{1}{99}-\frac{1}{100}$$

$$=-\frac{1}{8}+\frac{1}{101}=-\frac{93}{808}$$

**5)**

$$\sum_{r=1}^{n}\left(\frac{1}{r+8}-\frac{1}{r+7}\right)=$$

$$\frac{1}{9}+\frac{1}{10}+\frac{1}{11}+\cdots+\frac{1}{n+6}+\frac{1}{n+7}+\frac{1}{n+8}$$

$$-\frac{1}{8}-\frac{1}{9}-\frac{1}{10}-\cdots-\frac{1}{n+5}-\frac{1}{n+6}-\frac{1}{n+7}$$

$$=-\frac{1}{8}+\frac{1}{n+8}$$

**6)**

$$\sum_{r=1}^{n}\left(\frac{1}{r+3}-\frac{1}{r+4}\right)=$$

$$\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\cdots+\frac{1}{n+1}+\frac{1}{n+2}+\frac{1}{n+3}$$

$$-\frac{1}{5}-\frac{1}{6}-\frac{1}{7}-\cdots-\frac{1}{n+2}-\frac{1}{n+3}-\frac{1}{n+4}$$

$$=\frac{1}{4}-\frac{1}{n+4}$$

$$=\frac{n+4}{4(n+4)}-\frac{4}{4(n+4)}=\frac{n}{4(n+4)}\text{ as req.}$$

**7)**

$$\sum_{r=1}^{n}\left(\frac{1}{r+5}-\frac{1}{r+3}\right)=$$

$$\frac{1}{6}+\frac{1}{7}+\frac{1}{8}+\frac{1}{9}+\cdots+\frac{1}{n+3}+\frac{1}{n+4}+\frac{1}{n+5}$$

$$-\frac{1}{4}-\frac{1}{5}-\frac{1}{6}-\frac{1}{7}-\cdots-\frac{1}{n+1}-\frac{1}{n+2}-\frac{1}{n+3}$$

$$=-\frac{1}{4}-\frac{1}{5}+\frac{1}{n+4}+\frac{1}{n+5}$$

$$=-\frac{9}{20}+\frac{1}{n+4}+\frac{1}{n+5}$$

**8) a)**

$$\frac{1}{r+4}-\frac{1}{r+5}=\frac{r+5}{(r+4)(r+5)}-\frac{r+4}{(r+4)(r+5)}$$

$$=\frac{r+5-r-4}{(r+4)(r+5)}$$

$$=\frac{1}{(r+4)(r+5)} \text{ as req.}$$

**8) b)**

$$\sum_{r=1}^{n}\frac{1}{(r+4)(r+5)}=\sum_{r=1}^{n}\left(\frac{1}{r+4}-\frac{1}{r+5}\right)=$$

$$\begin{array}{ll}
\frac{1}{5} & -\frac{1}{6} \\
+\frac{1}{6} & -\frac{1}{7} \\
+\frac{1}{7} & -\frac{1}{8} \\
\ldots & \ldots \\
+\frac{1}{n+2} & -\frac{1}{n+3} \\
+\frac{1}{n+3} & -\frac{1}{n+4} \\
+\frac{1}{n+4} & -\frac{1}{n+5}
\end{array}$$

$$=\frac{1}{5}-\frac{1}{n+5}$$

**9) a)**

$$\frac{1}{r+1}-\frac{1}{r+2}=\frac{r+2}{(r+1)(r+2)}-\frac{r+1}{(r+1)(r+2)}$$
$$=\frac{r+2-r-1}{(r+1)(r+2)}$$
$$=\frac{1}{(r+1)(r+2)}$$

**9) b)**

$$\sum_{r=1}^{n}\frac{1}{(r+1)(r+2)}=\sum_{r=1}^{n}\left(\frac{1}{r+1}-\frac{1}{r+2}\right)=$$

$$\frac{1}{2}-\frac{1}{3}$$
$$+\frac{1}{3}-\frac{1}{4}$$
$$+\frac{1}{4}-\frac{1}{5}$$
$$\dots\quad\dots$$
$$+\frac{1}{n-1}-\frac{1}{n}$$
$$+\frac{1}{n}-\frac{1}{n+1}$$
$$+\frac{1}{n+1}-\frac{1}{n+2}$$

$$=\frac{1}{2}-\frac{1}{n+2}$$
$$=\frac{n+2}{2(n+2)}-\frac{2}{2(n+2)}=\frac{n}{2n+4}$$

**10) a)**

$\frac{1}{r-1}-\frac{1}{r+1}=\frac{r+1}{(r-1)(r+1)}-\frac{r-1}{(r-1)(r+1)}$

$=\frac{r+1-r+1}{(r-1)(r+1)}$

$=\frac{2}{(r-1)(r+1)}$ as req.

**10) b)**

$\frac{1}{r^2-1}=\frac{1}{(r-1)(r+1)}$

$$\sum_{r=5}^{50}\frac{1}{r^2-1}=\frac{1}{2}\sum_{r=5}^{50}\frac{2}{(r-1)(r+1)}=\frac{1}{2}\sum_{r=5}^{50}\left(\frac{1}{r-1}-\frac{1}{r+1}\right)$$

Since

$$\sum_{r=5}^{50}\left(\frac{1}{r-1}-\frac{1}{r+1}\right)=$$

$\frac{1}{4}$ $-\frac{1}{6}$

$+\frac{1}{5}$ $-\frac{1}{7}$

$+\frac{1}{6}$ $-\frac{1}{8}$

... ...

$+\frac{1}{47}$ $-\frac{1}{49}$

$+\frac{1}{48}$ $-\frac{1}{50}$

$+\frac{1}{49}$ $-\frac{1}{51}$

$=\frac{1}{4}+\frac{1}{5}-\frac{1}{50}-\frac{1}{51}$

$=\frac{2093}{5100}$

$$\frac{1}{2}\sum_{r=5}^{50}\left(\frac{1}{r-1}-\frac{1}{r+1}\right)=\frac{2093}{10200}$$

**11) a)**

$\frac{1}{r+3}-\frac{1}{r+4}=\frac{r+4}{(r+3)(r+4)}-\frac{r+3}{(r+3)(r+4)}=\frac{1}{(r+3)(r+4)}=\frac{1}{r^2+7r+12}$ as req.

**11) b)**

$$\sum_{r=8}^{22}\frac{1}{r^2+7r+12}=\sum_{r=8}^{22}\left(\frac{1}{r+3}-\frac{1}{r+4}\right)=$$

$$\begin{array}{ll}
\frac{1}{11} & -\frac{1}{12} \\
+\frac{1}{12} & -\frac{1}{13} \\
+\frac{1}{33} & -\frac{1}{14} \\
\ldots & \ldots \\
+\frac{1}{23} & -\frac{1}{24} \\
+\frac{1}{24} & -\frac{1}{25} \\
+\frac{1}{25} & -\frac{1}{26}
\end{array}$$

$=\frac{1}{11}-\frac{1}{26}=\frac{15}{286}$

**12) a)**

$\frac{1}{r-1}-\frac{1}{r+2}=\frac{r+2}{(r-1)(r+2)}-\frac{r-1}{(r-1)(r+2)}$

$=\frac{2--1}{(r-1)(r+2)}$

$=\frac{3}{(r-1)(r+2)}$ as req.

**12) b)**

$$\sum_{r=2}^{n}\frac{1}{(r-1)(r+2)}=\frac{1}{3}\sum_{r=2}^{1}\frac{3}{(r-1)(r+2)}=\frac{1}{3}\sum_{r=2}^{n}\left(\frac{1}{r-1}-\frac{1}{r+2}\right)$$

Since

$$\sum_{r=2}^{n}\left(\frac{1}{r-1}-\frac{1}{r+2}\right)=$$

| | |
|---|---|
| $1$ | $-\frac{1}{4}$ |
| $+\frac{1}{2}$ | $-\frac{1}{5}$ |
| $+\frac{1}{3}$ | $-\frac{1}{6}$ |
| $+\frac{1}{4}$ | $-\frac{1}{7}$ |
| ... | ... |
| $+\frac{1}{n-3}$ | $-\frac{1}{n}$ |
| $+\frac{1}{n-2}$ | $-\frac{1}{n+1}$ |
| $+\frac{1}{n-1}$ | $-\frac{1}{n+2}$ |

$=1+\frac{1}{2}+\frac{1}{3}-\frac{1}{n}-\frac{1}{n+1}-\frac{1}{n+2}$

$=\frac{11}{6}-\frac{1}{n}-\frac{1}{n+1}-\frac{1}{n+2}$

$$\frac{1}{3}\sum_{r=2}^{n}\left(\frac{1}{r-1}-\frac{1}{r+2}\right)=\frac{1}{3}\left(\frac{11}{6}-\frac{1}{n}-\frac{1}{n+1}-\frac{1}{n+2}\right)$$

**13)**

$$\frac{1}{r^3}-\frac{1}{(r+1)^3}=\frac{(r+1)^3}{r^3(r+1)^3}-\frac{r^3}{r^3(r+1)^3}$$

$$=\frac{(r+1)^3-r^3}{r^3(r+1)^3}$$

$$=\frac{r^3+3r^2+3r+1-r^3}{r^3(r+1)^3}$$

$$=\frac{3r^2+3r+1}{r^3(r+1)^3}$$

$$\therefore \sum_{r=1}^{n}\frac{3r^2+3r+1}{r^3(r+1)^3}=\sum_{r=1}^{n}\left(\frac{1}{r^3}-\frac{1}{(r+1)^3}\right)$$

$$\begin{array}{ll} 1 & -\frac{1}{2^3} \\ +\frac{1}{2^3} & -\frac{1}{3^3} \\ +\frac{1}{3^3} & -\frac{1}{4^3} \\ +\frac{1}{4^3} & -\frac{1}{5^3} \\ \cdots & \cdots \\ +\frac{1}{(n-2)^3} & -\frac{1}{(n-1)^2} \\ +\frac{1}{(n-1)^3} & -\frac{1}{n^3} \\ +\frac{1}{n^3} & -\frac{1}{(n+1)^3} \end{array}$$

$$=1-\frac{1}{(n+1)^3}$$

**14)**

$$\frac{1}{r-2}-\frac{1}{r}=\frac{r}{r(r-2)}-\frac{r-2}{r(r-2)}=\frac{2}{r(r-2)}$$

$$\therefore \sum_{r=3}^{n}\frac{1}{r^2-2r}=\frac{1}{2}\sum_{r=3}^{n}\frac{2}{r(r-2)}=\frac{1}{2}\sum_{r=3}^{n}\left(\frac{1}{r-2}-\frac{1}{r}\right)$$

Since

$$\sum_{r=3}^{n}\left(\frac{1}{r-2}-\frac{1}{r}\right)=$$

| | |
|---|---|
| $1$ | $-\frac{1}{3}$ |
| $+\frac{1}{2}$ | $-\frac{1}{4}$ |
| $+\frac{1}{3}$ | $-\frac{1}{5}$ |
| $+\frac{1}{4}$ | $-\frac{1}{6}$ |
| … | … |
| $+\frac{1}{n-4}$ | $-\frac{1}{n-2}$ |
| $+\frac{1}{n-3}$ | $-\frac{1}{n-1}$ |
| $+\frac{1}{n-2}$ | $-\frac{1}{n}$ |

$$=1+\frac{1}{2}-\frac{1}{n-1}-\frac{1}{n}$$

$$=\frac{3}{2}-\frac{1}{n-1}-\frac{1}{n}$$

$$\sum_{r=3}^{n}\frac{1}{r^2-2r}=\frac{1}{2}\left(\frac{3}{2}-\frac{1}{n-1}-\frac{1}{n}\right)$$

$$=\frac{1}{2}\left(\frac{3n(n-1)}{2n(n-1)}-\frac{2n}{2n(n-1)}-\frac{2(n-1)}{2n(n-1)}\right)$$

$$=\frac{1}{2}\left(\frac{3n^2-3n}{2n(n-1)}-\frac{2n}{2n(n-1)}-\frac{2n-2}{2n(n-1)}\right)$$

$$=\frac{1}{2}\left(\frac{3n^2-7n+2}{2n(n-1)}\right)$$

$$=\frac{(3n-1)(n-2)}{4n(n-1)}$$

**15) a)**

$$\frac{1}{2r+1}-\frac{1}{2r+3}=\frac{2r+3}{(2r+1)(2r+3)}-\frac{2r+1}{(2r+1)(2r+3)}$$
$$=\frac{2}{(2r+1)(2r+3)}$$

**15) b)**

$$\sum_{r=1}^{n}\frac{2}{(2r+1)(2r+3)}=\sum_{r=1}^{n}\left(\frac{1}{2r+1}-\frac{1}{2r+3}\right)=$$

$$\begin{array}{rr}
\frac{1}{3} & -\frac{1}{5} \\
+\frac{1}{5} & -\frac{1}{7} \\
+\frac{1}{7} & -\frac{1}{9} \\
+\frac{1}{9} & -\frac{1}{11} \\
\cdots & \cdots \\
+\frac{1}{2n-3} & -\frac{1}{2n-1} \\
+\frac{1}{2n-1} & -\frac{1}{2n+1} \\
+\frac{1}{2n+1} & -\frac{1}{2n+3}
\end{array}$$

$$=\frac{1}{3}-\frac{1}{2n+3}$$

So since there are 15 terms in the series, the sum is $\frac{1}{3}-\frac{1}{30+3}=\frac{10}{33}$

**16) a)**

$\frac{1}{r}-\frac{1}{r+1}=\frac{r+1}{r(r+1)}-\frac{r}{r(r+1)}=\frac{1}{r(r+1)}$

**16) b)**

$$\sum_{r=1}^{n}\frac{1}{r(r+1)}=\sum_{r=1}^{n}\left(\frac{1}{r}-\frac{1}{r+1}\right)=$$

$\frac{1}{1}$ $-\frac{1}{2}$

$+\frac{1}{2}$ $-\frac{1}{3}$

$+\frac{1}{3}$ $-\frac{1}{4}$

... ...

$+\frac{1}{n-2}$ $-\frac{1}{n-1}$

$+\frac{1}{n-1}$ $-\frac{1}{n}$

$+\frac{1}{n}$ $-\frac{1}{n+1}$

$=1-\frac{1}{n+1}$

So $1-\frac{1}{n+1}>0.98$

$n+1-1>0.98(n+1)$

$n>0.98n+0.98$

$0.02n>0.98$

$n>49$

So since it needs to exceed $0.98$ then $n=50$

So $50$ terms

**17) a)**

$$\frac{1}{2r}-\frac{1}{2r+2}=\frac{2r+2}{2r(2r+2)}-\frac{2r}{2r(2r+2)}=\frac{2}{2r(2r+2)}=\frac{1}{2r(r+1)}$$

**17) b)**

$$\sum_{r=1}^{n}\frac{1}{2r(r+1)}=\sum_{r=1}^{n}\left(\frac{1}{2r}-\frac{1}{2r+2}\right)=$$

$$\begin{array}{ll}
\frac{1}{2} & \\
+\frac{1}{4} & -\frac{1}{4} \\
+\frac{1}{6} & -\frac{1}{6} \\
\cdots & -\frac{1}{8} \\
 & \cdots \\
+\frac{1}{n-2} & -\frac{1}{2n-2} \\
+\frac{1}{2n-2} & -\frac{1}{2n} \\
+\frac{1}{2n} & -\frac{1}{2n+2}
\end{array}$$

$$=\frac{1}{2}-\frac{1}{2n+2}$$

**17) c)**

As $n\to\infty$, $\frac{1}{2n+2}\to 0$

$$\sum_{r=1}^{\infty}\frac{1}{2r(r+1)}=\frac{1}{2}-0=\frac{1}{2}$$

**18) a)**

$\frac{1}{2}[(r+1)(r+2)-r(r+1)]$

$=\frac{1}{2}(r+1)[r+2-r]$

$=\frac{1}{2}(r+1)\times 2$

$=(r+1)$ as req.

**18) b)**

$$\sum_{r=1}^{n} 2(r+1)=\sum_{r=1}^{n} 2\times\frac{1}{2}[(r+1)(r+2)-r(r+1)]$$

$$=\sum_{r=1}^{n}(r+1)(r+2)-r(r+1)=$$

$2\times 3$

$+3\times 4$

$+4\times 5$

…

$+(n-1)n$

$+n(n+1)$

$+(n+1)(n+2)$

$-1\times 2$

$-2\times 3$

$-3\times 4$

…

$-(n-1)(n+1)$

$-(n-1)n$

$-n(n+1)$

$=(n+1)(n+2)-1\times 2$

$=(n+1)(n+2)-2$

$=n^2+3n$

**19) a)**

$\frac{1}{3}[(r+1)(r+2)(r+3) - r(r+1)(r+2)]$

$= \frac{1}{3}(r+1)(r+2)[r+3-r]$

$= \frac{1}{3}(r+1)(r+2) \times 3$

$= (r+1)(r+2)$ as req.

**19) b)**

$$\sum_{r=1}^{n} 3(r+1)(r+2) = \sum_{r=1}^{n} 3 \times \frac{1}{3}[(r+1)(r+2)(r+3) - r(r+1)(r+2)]$$

$$= \sum_{r=1}^{n} (r+1)(r+2)(r+3) - r(r+1)(r+2) =$$

$\cancel{2 \times 3 \times 4}$ $\quad -1 \times 2 \times 3$

$\cancel{+3 \times 4 \times 5}$ $\quad \cancel{-2 \times 3 \times 4}$

$\cancel{+4 \times 5 \times 6}$ $\quad \cancel{-3 \times 4 \times 5}$

$\cancel{\ldots}$ $\quad \cancel{\ldots}$

$\cancel{+(n-1)n(n+1)}$ $\quad \cancel{-(n-1)(n+1)(n+2)}$

$\cancel{+n(n+1)(n+2)}$ $\quad \cancel{-(n-1)n(n+1)}$

$+(n+1)(n+2)(n+3)$ $\quad \cancel{-n(n+1)(n+2)}$

$= (n+1)(n+2)(n+3) - 1 \times 2 \times 3$

$= (n+1)(n+2)(n+3) - 6$

**20)**

$$\sum_{r=1}^{n}\left(\frac{2}{r^2}-\frac{2}{(r+1)^2}\right)=$$

$$\begin{array}{rr}
\frac{2}{1} & -\frac{2}{4} \\
+\frac{2}{4} & -\frac{2}{9} \\
+\frac{2}{9} & -\frac{2}{16} \\
\ldots & \ldots \\
+\frac{2}{(n-2)^2} & -\frac{2}{(n-1)^2} \\
+\frac{2}{(n-1)^2} & -\frac{2}{n^2} \\
+\frac{2}{n^2} & -\frac{2}{(n+1)^2}
\end{array}$$

$$=2-\frac{2}{(n+1)^2}$$

So

$$2-\frac{2}{(n+1)^2}>1.95$$

$$0.05>\frac{2}{(n+1)^2}$$

$$0.05(n+1)^2>2$$

$$0.05n^2+0.1n-1.95>0$$

$$n>5.32456\ldots\therefore n=6$$

**21) a)**

$\frac{1}{r!}-\frac{1}{(r+1)!}=\frac{r+1}{(r+1)r!}-\frac{1}{(r+1)r!}=\frac{r}{(r+1)r!}$ as req.

**21) b)**

$$\sum_{r=1}^{n}\frac{r}{(r+1)r!}=\sum_{r=1}^{n}\left(\frac{1}{r!}-\frac{1}{(r+1)!}\right)=$$

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\cdots+\frac{1}{(n-2)!}+\frac{1}{(n-1)!}+\frac{1}{n!}$$

$$-\frac{1}{2!}-\frac{1}{3!}-\frac{1}{4!}-\cdots-\frac{1}{(n-1)!}-\frac{1}{n!}-\frac{1}{(n+1)!}$$

$$=1-\frac{1}{(n+1)!}$$

**22) a)**

$(r+2)^3 - r^3 = r^3 + 6r^2 + 12r + 8 - r^3 = 6r^2 + 12r + 8$

**22) b)**

$$\sum_{r=1}^{n} 3r^2 + 6r + 4 = \frac{1}{2}\sum_{r=1}^{n}((r+2)^3 - r^3)$$

Since

$$\sum_{r=1}^{n}((r+2)^3 - r^3) =$$

| | |
|---|---|
| ~~$3^3$~~ | $-1^3$ |
| ~~$+4^3$~~ | $-2^3$ |
| ~~$+5^3$~~ | ~~$-3^3$~~ |
| ... | ... |
| ~~$+n^3$~~ | ~~$-(n-2)^3$~~ |
| $+(n+1)^3$ | ~~$-(n-1)^3$~~ |
| $+(n+2)^3$ | ~~$-n^3$~~ |

$= -1 - 8 + (n+1)^3 + (n+2)^3$

$= -9 + (n+1)^3 + (n+2)^3$

$$\therefore \sum_{r=1}^{n} 3r^2 + 6r + 4 = -\frac{9}{2} + \frac{1}{2}(n+1)^3 + \frac{1}{2}(n+2)^3$$

$$= -\frac{9}{2} + \frac{1}{2}(n^3 + 3n^2 + 3n + 1) + \frac{1}{2}(n^3 + 6n^2 + 12n + 8)$$

$$= -\frac{9}{2} + \frac{1}{2}(n^3 + 3n^2 + 3n + 1) + \frac{1}{2}(n^3 + 6n^2 + 12n + 8)$$

$$= n^3 + \frac{9}{2}n^2 + \frac{15}{2}n$$

$$= \frac{1}{2}n(2n^2 + 9n + 15)$$

**22) c)**

$$\sum_{r=1}^{n} r(r+2) = \sum_{r=1}^{n} r^2 + 2r = \frac{1}{3}\left(\sum_{r=1}^{n} 3r^2 + 6r + 4\right) - \frac{1}{3}\left(\sum_{r=1}^{n} 4\right)$$

$$= \frac{1}{3} \times \frac{1}{2} n(2n^2 + 9n + 15) - \frac{1}{3} \times 4n$$

$$= \frac{1}{6} n(2n^2 + 9n + 15 - 8)$$

$$= \frac{1}{6} n(2n^2 + 9n + 7)$$

$$= \frac{1}{6} n(n+1)(2n+7)$$

**23) a)**

$(2r+1)^3 - (2r-1)^3 = 8r^3 + 12r^2 + 6r + 1 - 8r^3 + 12r^2 - 6r + 1$

$= 24r^2 + 2$ as req.

**23) b)**

$$\sum_{r=1}^{n} 24r^2 + 2 = \sum_{r=1}^{n}((2r+1)^3 - (2r-1)^3) =$$

$3^3$ $\quad -1^3$

$+5^3$ $\quad -3^3$

$+7^3$ $\quad -5^3$

… $\quad$ …

$+(2n-3)^3$ $\quad -(2n-5)^3$

$+(2n-1)^3$ $\quad -(2n-3)^3$

$+(2n+1)^3$ $\quad -(2n-1)^3$

$= (2n+1)^3 - 1$

$= 8n^3 + 12n^2 + 6n + 1 - 1$

$= 8n^3 + 12n^2 + 6n$

$$\therefore \sum_{r=1}^{n} r^2 = \frac{1}{24}\sum_{r=1}^{n}(24r^2 + 2) - \frac{1}{24}\sum_{r=1}^{n} 2$$

$= \frac{1}{24}(8n^3 + 12n^2 + 6n) - \frac{1}{24}(2n)$

$= \frac{1}{24}n(8n^2 + 12n + 6 - 2)$

$= \frac{1}{6}n(2n^2 + 3n + 1)$

$= \frac{1}{6}n(2n+1)(n+1)$ as req.

**24) a)**

$(2r+1)^4 - (2r-1)^4 = 16r^4 + 32r^3 + 24r^2 + 8r + 1 - 16r^4 + 32r^3 - 24r^2 + 8r - 1$

$= 64r^3 + 16r$ as req.

**24) b)**

$$\sum_{r=1}^{n} 64r^3 + 16r = \sum_{r=1}^{n}((2r+1)^4 - (2r-1)^4) =$$

$3^4$ $\quad -1^4$

$+5^4$ $\quad -3^4$

$+7^4$ $\quad -5^4$

$\ldots$ $\quad \ldots$

$+(2n-3)^4$ $\quad -(2n-5)^4$

$+(2n-1)^4$ $\quad -(2n-3)^4$

$+(2n+1)^4$ $\quad -(2n-1)^4$

$= (2n+1)^4 - 1$

$= 16n^4 + 32n^3 + 24n^2 + 8n + 1 - 1$

$= 16n^4 + 32n^3 + 24n^2 + 8n$

$$\therefore \sum_{r=1}^{n} r^3 = \frac{1}{64}\sum_{r=1}^{n}(64r^3 + 16r) - \frac{1}{4}\sum_{r=1}^{n} r$$

$= \frac{1}{64}(16n^4 + 32n^3 + 24n^2 + 8n) - \frac{1}{4}\left(\frac{1}{2}n(n+1)\right)$

$= \frac{1}{8}(2n^4 + 4n^3 + 3n^2 + n) - \frac{1}{8}n(n+1)$

$= \frac{1}{8}n(2n^3 + 4n^2 + 3n + 1 - n - 1)$

$= \frac{1}{8}n(2n^3 + 4n^2 + 2n)$

$= \frac{1}{4}n^2(n^2 + 2n + 1)$

$= \frac{1}{4}n^2(n+1)^2$ as req.

**25) a)**

$(2r+1)^5 - (2r-1)^5$

$= 32r^5 + 80r^4 + 80r^3 + 40r^2 + 10r + 1 - 32r^5 + 80r^4 - 80r^3 + 40r^2 - 10r + 1$

$= 160r^4 + 80r^2 + 2$ as req.

**25) b)**

$$\sum_{r=1}^{n} 160r^4 + 80r^2 + 2 = \sum_{r=1}^{n}((2r+1)^5 - (2r-1)^5) =$$

$3^5$ $\quad -1^5$

$+5^5$ $\quad -3^5$

$+7^5$ $\quad -5^5$

$\ldots$ $\quad \ldots$

$+(2n-3)^5$ $\quad -(2n-5)^5$

$+(2n-1)^5$ $\quad -(2n-3)^5$

$+(2n+1)^5$ $\quad -(2n-1)^5$

$= (2n+1)^5 - 1$

$= 32n^5 + 80n^4 + 80n^3 + 40n^2 + 10n + 1 - 1$

$= 32n^5 + 80n^4 + 80n^3 + 40n^2 + 10n$

$$\therefore \sum_{r=1}^{n} r^4 = \frac{1}{160}\sum_{r=1}^{n}(160r^4 + 80r^2 + 2) - \frac{1}{2}\sum_{r=1}^{n} r^2 - \frac{1}{80}\sum_{r=1}^{n} 1$$

$$= \frac{1}{160}(32n^5 + 80n^4 + 80n^3 + 40n^2 + 10n) - \frac{1}{2}\left(\frac{1}{6}n(n+1)(2n+1)\right) - \frac{1}{80}n$$

$$= \frac{1}{80}n(16n^4 + 40n^3 + 40n^2 + 20n + 5) - \frac{1}{12}n(n+1)(2n+1) - \frac{1}{80}n$$

$$= n\left[\frac{1}{5}n^4 + \frac{1}{2}n^3 + \frac{1}{2}n^2 + \frac{1}{4}n + \frac{1}{16} - \frac{1}{6}n^2 - \frac{1}{4}n - \frac{1}{12} - \frac{1}{80}\right]$$

$$= n\left[\frac{1}{5}n^4 + \frac{1}{2}n^3 + \frac{1}{3}n^2 - \frac{1}{30}\right]$$

$$= \frac{1}{30}n[6n^4 + 15n^3 + 10n^2 - 1]$$

$$= \frac{1}{30}n(n+1)[6n^3 + 9n^2 + n - 1]$$

$$= \frac{1}{30}n(n+1)(2n+1)(3n^2 + 3n - 1)$$ as req.

**26)**

$$\sum_{r=k}^{100}\left(\frac{2}{r}-\frac{2}{r+1}\right)=$$

$\frac{2}{k}$ $-\frac{2}{k+1}$

$+\frac{2}{k+1}$ $-\frac{2}{k+2}$

$+\frac{2}{k+2}$ $-\frac{2}{k+3}$

... ...

$+\frac{2}{98}$ $-\frac{2}{99}$

$+\frac{2}{99}$ $-\frac{2}{100}$

$+\frac{2}{100}$ $-\frac{2}{101}$

$=\frac{2}{k}-\frac{2}{101}$

$\therefore \frac{2}{k}-\frac{2}{101}=\frac{65}{1818}$

$\frac{2}{k}=\frac{1}{18}$

$\therefore k=36$

**27) a)**

$$\frac{2}{r}-\frac{1}{r+1}-\frac{1}{r+2}=\frac{2(r+1)(r+2)}{r(r+1)(r+2)}-\frac{r(r+2)}{r(r+1)(r+2)}-\frac{r(r+1)}{r(r+1)(r+2)}$$

$$=\frac{2r^2+6r+4}{r(r+1)(r+2)}-\frac{r^2+2r}{r(r+1)(r+2)}-\frac{r^2+r}{r(r+1)(r+2)}$$

$=\frac{3r+4}{r(r+1)(r+2)}$ as req.

**27) b)**

$$\sum_{r=1}^{n}\frac{3r+4}{r(r+1)(r+2)}=\sum_{r=1}^{n}\left(\frac{2}{r}-\frac{1}{r+1}-\frac{1}{r+2}\right)=$$

| | | |
|---|---|---|
| $\frac{2}{1}$ | $-\frac{1}{2}$ | $-\frac{1}{3}$ |
| $+\frac{2}{2}$ | $-\frac{1}{3}$ | $-\frac{1}{4}$ |
| $+\frac{2}{3}$ | $-\frac{1}{4}$ | $-\frac{1}{5}$ |
| ... | ... | ... |
| $+\frac{2}{n-2}$ | $-\frac{1}{n-1}$ | $-\frac{1}{n}$ |
| $+\frac{2}{n-1}$ | $-\frac{1}{n}$ | $-\frac{1}{n+1}$ |
| $+\frac{2}{n}$ | $-\frac{1}{n+1}$ | $-\frac{1}{n+2}$ |

$$=2-\frac{1}{2}+\frac{2}{2}-\frac{1}{n+1}-\frac{1}{n+1}-\frac{1}{n+2}$$

$$=\frac{5}{2}-\frac{2}{n+1}-\frac{1}{n+2}$$

**27) c)**

$$\sum_{r=k}^{25}\frac{3r+4}{r(r+1)(r+2)}=\sum_{r=k}^{25}\left(\frac{2}{r}-\frac{1}{r+1}-\frac{1}{r+2}\right)=$$

| | | |
|---|---|---|
| $\frac{2}{k}$ | $-\frac{1}{k+1}$ | $-\frac{1}{k+2}$ |
| $+\frac{2}{k+1}$ | $-\frac{1}{k+2}$ | $-\frac{1}{k+3}$ |
| $+\frac{2}{k+2}$ | $-\frac{1}{k+3}$ | $-\frac{1}{k+4}$ |
| ... | ... | ... |
| $+\frac{2}{23}$ | $-\frac{1}{24}$ | $-\frac{1}{25}$ |
| $+\frac{2}{24}$ | $-\frac{1}{25}$ | $-\frac{1}{26}$ |
| $+\frac{2}{25}$ | $-\frac{1}{26}$ | $-\frac{1}{27}$ |

$=\frac{2}{k}-\frac{1}{k+1}+\frac{2}{k+1}-\frac{1}{26}-\frac{1}{26}-\frac{1}{27}$

$=\frac{2}{k}+\frac{1}{k+1}-\frac{40}{351}$

$\therefore \frac{2}{k}+\frac{1}{k+1}-\frac{40}{351}=\frac{1174}{12\,285}$

$\frac{2(k+1)}{k(k+1)}+\frac{k}{k(k+1)}=\frac{22}{105}$

$2k+2+k=\frac{22}{105}k(k+1)$

$315k+210=22k^2+22k$

$22k^2-293k-210=0$

which has solutions $k=-\frac{15}{22}$ and $k=14$

$\therefore k=14$

**28) a)**

$2^{r+1}(r-1)-2^r(r-2)=2\times 2^r(r-1)-2^r(r-2)$

$=2\times 2^r\times r-2\times 2^r-2^r\times r+2\times 2^r$

$=2\times 2^r\times r-2^r\times r$

$=r2^r$ as req.

**28) b)**

$$\sum_{r=1}^{n} r2^r = \sum_{r=1}^{n} \left(2^{r+1}(r-1) - 2^r(r-2)\right) =$$

| | |
|---|---|
| $2^2 \times 0$ | $-2^1 \times -1$ |
| $+2^3 \times 1$ | $-2^2 \times 0$ |
| $+2^4 \times 2$ | $-2^3 \times 1$ |
| … | … |
| $+2^{n-1}(n-3)$ | $-2^{n-2}(n-4)$ |
| $+2^n(n-2)$ | $-2^{n-1}(n-3)$ |
| $+2^{n+1}(n-1)$ | $-2^n(n-2)$ |

$= 2 + 2^{n+1}(n-1)$ as req.

**29) a)**

$$\ln\left(1 + \frac{1}{r+1}\right) = \ln\left(\frac{r+1}{r+1} + \frac{1}{r+1}\right) = \ln\left(\frac{r+2}{r+1}\right) = \ln(r+2) - \ln(r+1)$$

**29) b)**

$$\sum_{r=1}^{n} \ln\left(1+\frac{1}{r+1}\right) = \sum_{r=1}^{n} (\ln(r+2) - \ln(r+1)) =$$

| | |
|---|---|
| $\ln 3$ | $-\ln 2$ |
| $+\ln 4$ | $-\ln 3$ |
| $+\ln 5$ | $-\ln 4$ |
| ... | ... |
| $+\ln(n)$ | $-\ln(n-1)$ |
| $+\ln(n+1)$ | $-\ln n$ |
| $+\ln(n+2)$ | $-\ln(n+1)$ |

$= \ln(n+2) - \ln 2$

$= \ln\left(\frac{n+2}{2}\right)$

**30)**

$$\sum_{r=1}^{n} \ln\left(\frac{r+1}{r}\right) = \sum_{r=1}^{n} (\ln(r+1) - \ln(r)) =$$

| | |
|---|---|
| $\ln 2$ | $-\ln 1$ |
| $+\ln 3$ | $-\ln 2$ |
| $+\ln 4$ | $-\ln 3$ |
| ... | ... |
| $+\ln(n-1)$ | $-\ln(n-2)$ |
| $+\ln(n)$ | $-\ln(n-1)$ |
| $+\ln(n+1)$ | $-\ln n$ |

$= \ln(n+1) - \ln 1 = \ln(n+1)$

**31)**

$$\sum_{r=21}^{50} \ln\left(\frac{r+1}{r-1}\right) = \sum_{r=21}^{50} (\ln(r+1) - \ln(r-1)) =$$

| | |
|---|---|
| ~~$\ln 22$~~ | $-\ln 20$ |
| ~~$+\ln 23$~~ | $-\ln 21$ |
| ~~$+\ln 24$~~ | ~~$-\ln 22$~~ |
| ... | ... |
| ~~$+\ln 49$~~ | ~~$-\ln 47$~~ |
| $+\ln 50$ | ~~$-\ln 48$~~ |
| $+\ln 51$ | ~~$-\ln 49$~~ |

$$= \ln 50 + \ln 51 - \ln 20 - \ln 21 = \ln\left(\frac{85}{14}\right)$$

**32) a)**

$$\sum_{r=1}^{n} \ln\left(\frac{r+3}{r+1}\right) = \sum_{r=1}^{n} (\ln(r+3) - \ln(r+1)) =$$

| | |
|---|---|
| ~~$\ln 4$~~ | $-\ln 2$ |
| ~~$+\ln 5$~~ | $-\ln 3$ |
| ~~$+\ln 6$~~ | ~~$-\ln 4$~~ |
| ... | ... |
| ~~$+\ln(n+1)$~~ | ~~$-\ln(n-1)$~~ |
| $+\ln(n+2)$ | ~~$-\ln n$~~ |
| $+\ln(n+3)$ | ~~$-\ln(n+1)$~~ |

$= \ln(n+2) + \ln(n+3) - \ln 2 - \ln 3$

$= \ln\big((n+2)(n+3)\big) - \ln 6$

$= \ln\left(\frac{1}{6}(n+2)(n+3)\right)$

**32) b)**

$$\sum_{r=1}^{n} \ln\left(\frac{r+3}{r+1}\right)^{36} = \sum_{r=1}^{n} 36\ln\left(\frac{r+3}{r+1}\right) = 36\ln\left(\frac{1}{6}(n+2)(n+3)\right)$$

**33) a)**

$$\sum_{r=1}^{n} \ln\left(\frac{2r+3}{2r+1}\right) = \sum_{r=1}^{n} (\ln(2r+3) - \ln(2r+1)) =$$

| | |
|---|---|
| $\ln 5$ | $-\ln 3$ |
| $+\ln 7$ | $-\ln 5$ |
| $+\ln 9$ | $-\ln 7$ |
| ... | ... |
| $+\ln(2n-1)$ | $-\ln(2n-3)$ |
| $+\ln(2n+1)$ | $-\ln(2n-1)$ |
| $+\ln(2n+3)$ | $-\ln(2n+1)$ |

$= \ln(2n+3) - \ln 3$

$= \ln\left(\frac{1}{3}(2n+3)\right)$

**33) b)**

$\ln\left(\frac{1}{3}(2n+3)\right) > 4$

$\frac{1}{3}(2n+3) > e^4$

$2n+3 > 3e^4$

$n > \frac{1}{2}(3e^4 - 3) \therefore n = 81$

**34)**

$$\sum_{r=6}^{n}\left(\frac{1}{r+1}-\frac{1}{r+3}\right)=$$

$$\frac{1}{7}$$
$$+\frac{1}{8}$$
$$+\frac{1}{9}$$
$$\cdots$$
$$+\frac{1}{n-1}$$
$$+\frac{1}{n}$$
$$+\frac{1}{n+1}$$

$$-\frac{1}{9}$$
$$-\frac{1}{10}$$
$$-\frac{1}{11}$$
$$\cdots$$
$$-\frac{1}{n+1}$$
$$-\frac{1}{n+2}$$
$$-\frac{1}{n+3}$$

$$=\frac{1}{7}+\frac{1}{8}-\frac{1}{n+2}-\frac{1}{n+3}$$

As $n\to\infty, \frac{1}{n+2}\to 0, \frac{1}{n+3}\to 0$

$$\therefore \sum_{r=6}^{\infty}\left(\frac{1}{r+1}-\frac{1}{r+3}\right)=\frac{1}{7}+\frac{1}{8}=\frac{15}{56}$$

**35) a)**

$\frac{1}{2(r+2)} - \frac{1}{2(r+4)} = \frac{r+4}{2(r+2)(r+4)} - \frac{r+2}{2(r+2)(r+4)} = \frac{2}{2(r+2)(r+4)} = \frac{1}{(r+2)(r+4)}$ as req.

**35) b)**

$$\sum_{r=1}^{n} \frac{1}{(r+2)(r+4)} = \sum_{r=1}^{n} \left(\frac{1}{2(r+2)} - \frac{1}{2(r+4)}\right) =$$

$$\frac{1}{2\times3} - \frac{1}{2\times5}$$
$$+\frac{1}{2\times4} - \frac{1}{2\times6}$$
$$+\frac{1}{2\times5} - \frac{1}{2\times7}$$
$$\ldots \quad \ldots$$
$$+\frac{1}{2n} - \frac{1}{2(n+2)}$$
$$+\frac{1}{2(n+1)} - \frac{1}{2(n+3)}$$
$$+\frac{1}{2(n+2)} - \frac{1}{2(n+4)}$$

$= \frac{7}{24} - \frac{1}{2(n+3)} - \frac{1}{2(n+4)}$

$\frac{7}{24} - \left(\frac{7}{24} - \frac{1}{2(n+3)} - \frac{1}{2(n+4)}\right) < \frac{19}{180}$

$\frac{1}{2(n+3)} + \frac{1}{2(n+4)} < \frac{19}{180}$

$\frac{n+4}{2(n+3)(n+4)} + \frac{n+3}{2(n+3)(n+4)} < \frac{19}{180}$

$\frac{2n+7}{(n+3)(n+4)} < \frac{19}{90}$

$180n + 630 < 19(n+3)(n+4)$

$180n + 630 < 19n^2 + 133n + 228$

$19n^2 - 47n - 402 > 0$

$n < -\frac{67}{19}$ or $n > 6$

$n$ is a positive integer number so $n \geq 7$ as req.